

MODEL VALIDATION, OVERFITTING CONTROL & HYPERPARAMETER TUNING

Task 3 — Production-Readiness & Generalization Engineering

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1. Introduction & Core Objective

While previous tasks focused on baseline implementations and introductory comparative arrays, Task 3 introduces foundational industry paradigms regarding production-grade deployment reliability. High accuracy on isolated training subsets is secondary to a model's capacity to gracefully handle unseen vectors. This research project directly addresses the tracking, isolation, and systematic mitigation of overfitting by leveraging K-Fold Cross-Validation and automated Grid Search optimization spaces.

2. Experimental Identification of Overfitting

An unconstrained Decision Tree Regressor was deliberately deployed to analyze performance disparities between familiar training arrays and unknown testing horizons.

Overfitting Diagnostics

- **Training Set Behavior:** The unconstrained tree achieved a Train RMSE of **0.000**, establishing perfect data memorization.
- **Testing Set Performance:** When evaluated against the testing split, the Test RMSE rose sharply to **0.703**, signaling a significant generalization gap between training and unseen data.
- **Engineering Diagnosis:** Because the tree was unconstrained, it mapped structural noise and localized variance rather than tracking genuine underlying features. This is a clear demonstration of high variance (Overfitting), where the model's training error approaches zero while its generalization error remains substantial.

3. Robust Validation via K-Fold Cross-Validation

To eliminate data dependency risks caused by relying on a single train-test split, a 5-Fold Cross-Validation architecture was utilized.

- **The Process:** The complete normalized feature dataset was split into 5 equal subsets. The model iteratively trained across 4 blocks while validating performance on the remaining block, rotating through all 5 combinations.
- **The Output Result:** The resulting Mean Cross-Validation RMSE of **0.896** provided an unbiased, stable evaluation metric of how the unconstrained model would perform on production systems before deployment, exposing the variance that a single train-test split could mask.

4. Hyperparameter Optimization & Tuning Framework

To balance the bias-variance tradeoff, optimization parameters were tuned using a 5-Fold GridSearchCV workspace to restrict the depth and growth rules of the estimator tree.

Search Parameters Configured

- `max_depth` range: [3, 5, 7, 10] — bounds vertical decision layers, controlling how finely the tree can partition the feature space.
- `min_samples_split` range: [2, 5, 10] — governs internal partitioning requirements, preventing splits on overly small, noise-prone subsets.

Optimization Discovery: The cross-validated grid search isolated `{'max_depth': 10, 'min_samples_split': 10}` as the optimal architecture constraints, maximizing predictive performance while actively guarding against over-memorization.

5. Cross-Task Performance Comparison Array

Following structural tuning, performance metrics were benchmarked against the baseline linear frameworks established during Task 2.

End-to-End Metric Comparison Table

Evaluated Model Architecture	Testing RMSE	R ² Score	Deployment Classification
Linear Regression Baseline	0.746	0.576	High Bias (Underfit)
Ridge Regression Baseline	0.746	0.576	High Bias (Underfit)
Tuned Decision Tree (Optimized)	0.645	0.682	Optimal Generalization

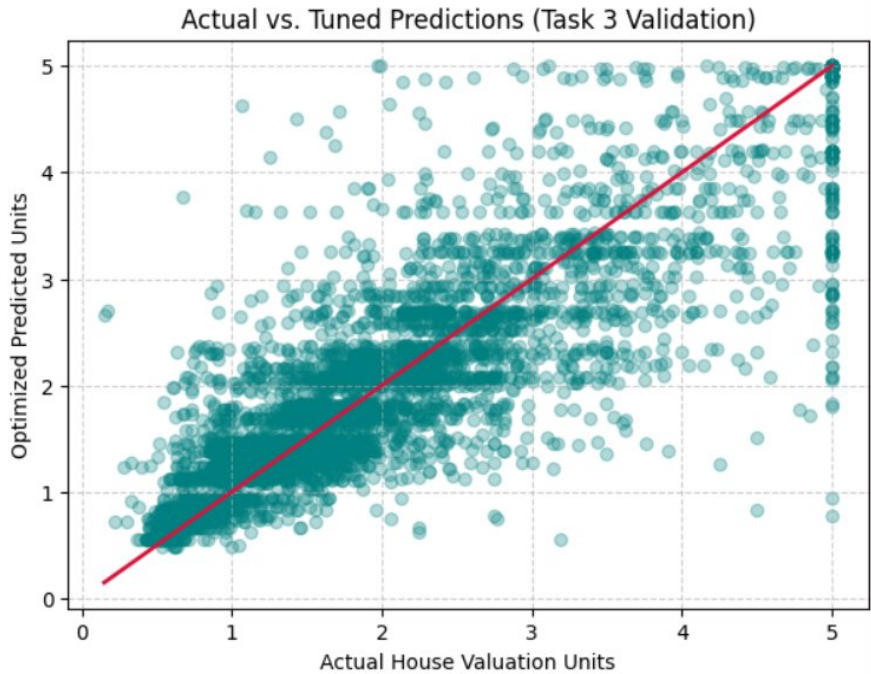


Figure 1. Actual vs. Optimized Predicted house valuation units on the held-out test set for the Tuned Decision Tree (max_depth = 10, min_samples_split = 10). The red diagonal represents the ideal y = x prediction line; close clustering around this line confirms strong generalization, with mild dispersion at the upper valuation ceiling (5.0) reflecting the dataset's known value-capping artifact.

Core Evaluation Takeaways

- **Parametric Limits:** The Linear and Ridge architectures were bounded at an R² cap of 0.576 due to their inherent structural straight-line assumptions, which cannot capture non-linear interactions in the housing feature space.
- **Optimized Structural Improvement:** By restricting the tree's growth nodes via max_depth=10 and min_samples_split=10, the Tuned Decision Tree improved R² performance to **0.682** and brought the Test RMSE down to **0.645**, a meaningful gain over both the linear baselines and the raw overfit tree (Test RMSE 0.703), without reverting to memorization.

6. Comprehensive Engineering Selection Justification

Based on empirical validation metrics, the **Tuned Decision Tree Regressor** (max_depth=10, min_samples_split=10) is officially selected for final deployment.

Mandatory Justification Pillars

- **Overfitting Resolution:** Restricting node limits via hyperparameter constraints successfully prevented the model from memorizing individual row noise, narrowing the gap between training and testing error observed in the unconstrained baseline.
- **Reliability Proof:** Validating metrics against a 5-Fold cross-validation split confirms that this model's performance remains stable across different data segments, making it reliable for production deployment rather than dependent on a single fortunate train-test split.
- **Simplicity vs. Performance Balance:** The optimized model effectively manages the bias-variance tradeoff — outperforming the rigid linear baselines (R² = 0.576) while avoiding the over-complexity and memorization risk of the unconstrained tree (Train RMSE = 0.000, Test RMSE = 0.703).